T34 첨단 패키징 소재의 기능별 요구와 개발 방향

LG Chem · 이광주 · KwangJoo Lee

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Introduction of LG Chem

Beyond a chemical company, LG Chem will become a leading science-oriented corporation that creates value for our customers.

Financial Results
*Included Subsidiaries

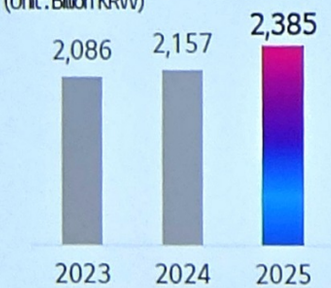
Sales in 2025
KRW **45.9**trn
(Approx.)

Workforce
17,389(Person)
Domestic 12,943 / Overseas 4,446

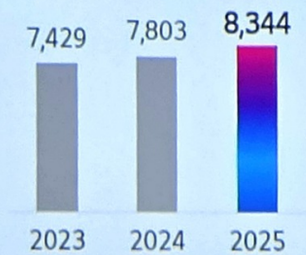
Sites of business
63
Domestic 15 / Overseas 48

R&D Status
*Included Subsidiaries

R&D Expenses
(Unit : Billion KRW)



R&D Workforce
(Unit : Person)



Business Area

Petrochemicals
- NCC/Polyolefins
- PVC/Plasticizers
- ABS
- Acrylates / SAP
- HPM(High Performance Materials)

Advanced Materials
- Electronics Materials
  - Cathode Materials
  - Engineering Materials
  - Battery Separator

Life Sciences
- Primary Care
- Specialty Care
- AVEO Oncology

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Introduction of Electronics Materials Division | Business Area

We have established strong partnerships with global customers across various industries.

Business Area

Industries	✓ Semiconductor	e-Mobility	Display
Products	**Packaging Materials** · Copper Clad Laminate · Die Attach Film · Non Conductive Film · Build Up Film · Photo Imageable Dielectric **Process Materials** · Back Grinding Tape · Mount Tape · Stripper	**Optical Solutions** · Switchable Glazing Film · Transparent Antenna Film **Adhesives Solutions** · Body & Paints · Electrical Component -e-Motor & Battery -Sensors & Infotainments	**OLED Materials** · Emitting Layer · Common Layer **Organic Insulating Materials** · Pixel Define Layer **OLED Encapsulation** · Thin Film Encapsulation Ink
Applications	Telecommunication	e-Motor	OLED TV

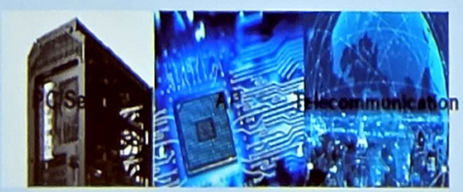




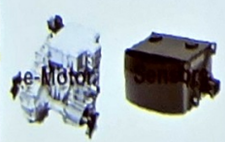
















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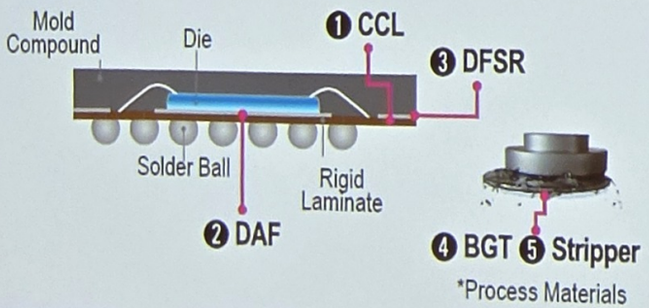
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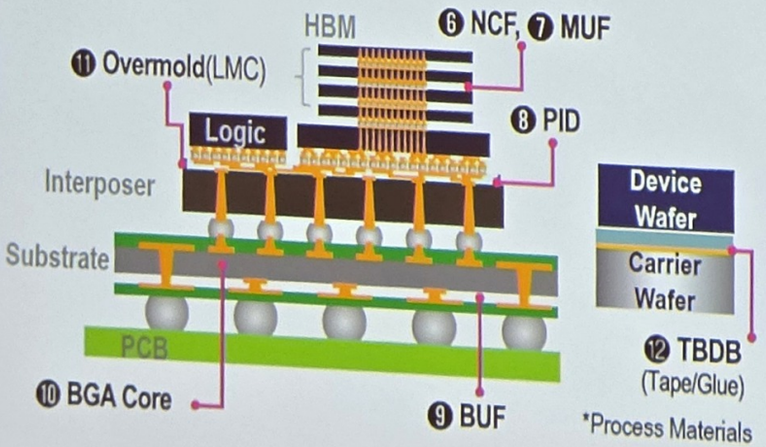
Introduction of Electronics Materials Division | Technologies

We are expanding Portfolio from Legacy to Advanced PKG to provide total PKG material solutions

Conventional Package



Advanced Package : 2.5D / 3D



Product	Function	Status
❶ CCL	✓ Copper Clad Laminate - Package substrate for connecting chip and PCB	HVM
❷ DAF	✓ Die Attach Film - Film-type adhesive for die-to-die or die-to-substrate bonding	HVM
❸ DFSR	✓ Dry Film Solder Resist - Film-type solder resist designed to protect the outermost circuit wiring of semiconductor package substrates	HVM
❹ BGT	✓ Back Grinding Tape - Protective tape for wafers during back grinding process	HVM
❺ Stripper	✓ Stripper - Photoresist stripping and substrate surface cleaning	HVM
❻ NCF	✓ Non-Conductive Film - Insulating adhesive film for chip-to-chip bonding and filling in HBM stacking	Under Development
❼ MUF	✓ Molded UnderFill - Fill the gap between chip and chip of HBM, under the flip-chip and encapsulate the entire die in a single step simultaneously	Under Development
❽ PID (Liquidus/Film)	✓ Photo Imageable Dielectric - Photosensitive liquid dielectric for insulation and fine via patterning	Under Development
❾ BUF	✓ Build-Up Film - Insulating film for sequential layer-by-layer stack formation in multilayer substrates	Under Development
❿ BGA Core	✓ Ball Grid Array Core - Core substrate for chip interconnection in BGA packages	Under Development
⓫ Overmold (LMC)	✓ Overmold(Liquid Molding Compound) - Liquidus type epoxy molding compound	Under Development
⓬ TBDB (Tape/Glue)	✓ Temporary Bonding De-Bonding Adhesive - Thin wafer handling bonding debonding or mounting adhesive	Under Development

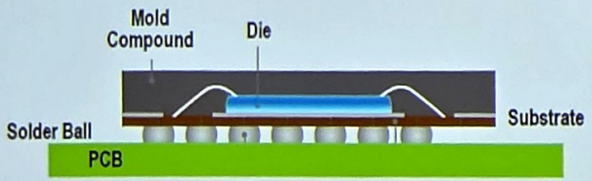
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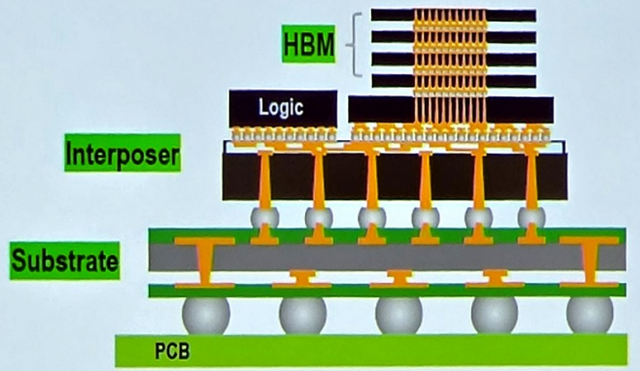
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Key Challenges of Advanced Packaging Materials

Conventional Package



Advanced Package : 2.5D / 3D



Fine Pitch

- PID (RDL)/PR
- Build-up
- Fine-pitch substrate
- Thin Copper
- Cu Plating Solution

High Thermal

- High K EMC
- **High K NCF**/MUF
- **High K DAF**
- TIM
- Cooling

High Speed (Low Dk/Df)

- Build-up Film
- PID (RDL)
- CCL/PPG, G/F
- Glass Core/Interposer
- CPO

Thin Wafer/ Sub. Handling

- WSS (TBDB)
- High Stiffness Carrier
- High Thermal Resistance Film

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